

JACOB G. HUNTER

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Research and Teaching Interests

Engineering Design, Cyber Physical Systems, Product Development, Human Factors,
Human Automation Interaction, Human-AI Teaming, Design Theory and Methodology

Education

Purdue University Ph.D. in Mechanical Engineering	West Lafayette, IN December 2025
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Purdue University M.S. in Mechanical Engineering	West Lafayette, IN May 2023
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Brigham Young University B.S. in Mechanical Engineering Secondary Major in French Studies	Provo, UT April 2020
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Professional Experience

Pennsylvania State University Research Scholar	University Park, PA February 2026 – Present
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Purdue University Graduate Research Assistant Instructor of Record, ME-20000	West Lafayette, IN June 2020 – December 2025 August – December 2024, 2025
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Honda Research Institute Student Associate	San Jose, CA August 2021 – December 2021
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Brigham Young University Undergraduate Research Assistant	Provo, UT August 2018 – June 2020
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French National Institute of Health & Medical Research Research & Development Intern	Paris, France May 2019 – June 2019
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Honors and Awards

Perry Graduate Teaching Fellowship, Thermodynamics I (ME 20000), School of Mechanical Engineering, Purdue University. August – December 2025.

Anil K. Bajaj Outstanding Service Scholarship, School of Mechanical Engineering, Purdue University. This scholarship recognizes mechanical engineering graduate students who provide outstanding service to the ME graduate student community, December 2024.

Ward A. Lambert Graduate Teaching Fellowship, Thermodynamics I (ME 20000), School of Mechanical Engineering, Purdue University. Served as an apprentice to a faculty mentor during the spring 2024 semester and lectured a section during the fall 2024 semester, January – May, August – December 2024.

Professional Grant Recipient, Purdue Graduate Student Government. Provided funding to attend the 7th SIG Tutorial on Design Theory in Paris, France, January 2023.

Top 10 Abstract, ASME Design Theory & Methodology Poster Session, August 2022.

Three Minute Thesis Finalist, Purdue University, April 2022.

First Place Elevator Pitch, Purdue University Graduate School, February 2021.

First Place, ASME Student Mechanisms & Robotics Design Competition, August 2020.

Elisabeth M. and Winchell M. Parsons Scholarship, ASME, September 2020.

Weidman Global Scholar, BYU Engineering, August 2019 – April 2020.

Student Design Essay Winner, Joint NSF & ASME Competition, August 2018.

Academic Scholarship, Brigham Young University, September 2016 – July 2019.

Publications

Journal Articles and Book Chapters

1. Hunter, Jacob G., Schonlau, S.L., Jain, N., and Reid, T. “A Methodology for Objective Design Fixation Analysis in Constrained Design Tools.” *Journal of Mechanical Design*, 2026, *Accepted*.
2. Choi, Joonwon, Hunter, J.G., Jain, N., Hwang, I., and Reid, T. “Distraction-aware Data-driven Hybrid Safety Analysis of Cyber-Physical-Human Systems Using Functional Near-infrared Spectroscopy.” *Journal of Aerospace Information Systems*, 2025, *Under Review*.
3. Hunter, Jacob G., Ulwelling, E., Konishi, M., Michelini, N., Modali, A., Mendoza, A., ... and Reid, T. “The Future of Mobility-as-a-Service: Trust Transfer Across Automated Mobilities, From Road to Sidewalk.” *Frontiers in Psychology*, 2023, vol. 14, pp. 17-37.
4. Jain, N., Reid, T., Akash, K., Yuh, M., and Hunter, J. (2023). Enabling Human-Aware Autonomy through Cognitive Modeling and Feedback Control. *Cyber-Physical-Human Systems: Fundamentals and Applications*, 91, (ed. A. Annaswamy et al.), UK: Wiley.

Patents

1. Sheffield, J.L., Lytle, A., Hunter, J.G., Hyatt, L., and Howell, L.L. “Integrated Cleaning Device and System for Optical Instruments.” U.S. Patent Application 18/000,804, filed August 3, 2023.

Conference Papers

1. J.G. Hunter, S.L. Schonlau, N. Jain, and T. Reid, “When Do Designers Choose AI Assistance? How Design Fixation Influences Requests for Help,” ASME 2026 International Design Engineering Technical Conferences & Computers and Information in Engineering Conference, Houston, TX, IDETC2026-195033, August 23–26, 2026. *Under Review*.

2. A.T. Beris, J.G. Hunter, and T. Reid, “Vibe Designing: Performance of Current Large Language Models on Engineering Truss Design,” ASME 2026 International Design Engineering Technical Conferences & Computers and Information in Engineering Conference, Houston, TX, IDETC2026-188679, August 23–26, 2026. *Under Review*.
3. J.R. Burton, S.A. McKinnon, J.G. Hunter, J.L. Salmon, and C.A. Mattson, “Predictive Models of Human Behavior in Engineering Design,” ASME 2026 International Design Engineering Technical Conferences & Computers and Information in Engineering Conference, Houston, TX, IDETC2026-194237, August 23–26, 2026. *Under Review*.
4. J. Choi, J.G. Hunter, I. Hwang, and T. Reid, “Distraction-aware Data-driven Hybrid Reachability Analysis of Cyber-Physical-Human Systems Using Functional Near-infrared Spectroscopy,” AIAA SCITECH 2025 Forum, Orlando, FL, January 6-10, 2025.
5. K.R. Ortiz, J.G. Hunter, A.J. Thorpe, M. Yuh, T. Reid, N. Jain, and M. Oishi, “Assessing the Relationship Between Learning Stages and Prefrontal Cortex Activation in a Psychomotor Task,” Proceedings of the Human Factors and Ergonomics Society 68th International Annual Meeting, Atlanta, GA, September 9–13, 2024.
6. S. Mehrotra, J.G. Hunter, M. Konishi, K. Akash, Z. Zheng, T. Misu, A. Kumar, T. Reid, and N. Jain, “Trust in Shared Automated Vehicles: Study on Two Mobility Platforms,” 103rd Transportation Research Board Annual Meeting, Washington D.C., January 8–12, 2023, arxiv.org/pdf/2303.09711.pdf.
7. J.G. Hunter, M. Konishi, N. Jain, K. Akash, X. Wu, T. Misu, and T. Reid, “The interaction gap: a step toward understanding trust in autonomous vehicles between encounters,” Proceedings of the Human Factors and Ergonomics Society 66th International Annual Meeting, Atlanta, GA, October 10–14, 2022.
8. M. Konishi, J.G. Hunter, Z.K. Zheng, T. Misu, K. Akash, T. Reid, and N. Jain, “Inferring Takeover in SAE Level 2 Automated Vehicles Using Driver-Based Behavioral and Psychophysiological Signals,” IFAC-PapersOnLine, 2022, vol. 55, iss. 41, pp. 7-12.
9. K. Ray, B. Cappuccilli, J. Hunter, M. Eichenauer, B.E. Sells, and T. Reid, “Ethical Human Centered Design: Practical Approaches for Aerospace Engineering Education,” AIAA Aviation Forum, Chicago, IL, IDETC2019-98419, June 27 – July 1, 2022, DOI:10.2514/6.2022-3711.
10. J.G. Hunter, C.A. Mattson, and S.P. Magleby, “Benefits of a Short-Term Engineering Study Abroad: A Survey of Students Over the Past 15 Years,” ASME 2019 International Design Engineering Technical Conferences & Computers and Information in Engineering Conference, Anaheim, CA, IDETC2019-98419, August 18–21, 2019.

Talks and Presentations

Poster Presentation, “Objective Design Fixation Analysis as a Basis for AI-Designer Teaming in Constrained Design,” NSF AI-EDSE Workshop, West Lafayette, Indiana, October 16, 2025.

Technical Presentation, “Studying Design Fixation Through Computer-Based Constrained Design,” ASME 2025 International Design Engineering Technical Conferences & Computers and Information Engineering Conference, Anaheim, California, August 17-20, 2025.

Workshop Presentation, “Understanding Neurocognitive Activity to Facilitate Human-Machine Teaming During a Drone Design Task,” 3rd Workshop on Trends in Human-AI Teaming for Engineering Design, Washington, DC, August, 25, 2024.

Invited Talk, “Your Time to Shine,” Mechanical Engineering Graduate Recruitment Visitation, Purdue University, West Lafayette, Indiana, February 23, 2024.

Technical Presentation, “Design Neurocognition in the First Stage of Product Development,” ASME 2023 International Design Engineering Technical Conferences & Computers and Information Engineering Conference, Boston, Massachusetts, August 20-23, 2023.

Panelist, Mechanical Engineering Graduate Recruitment Visitation, Purdue University, West Lafayette, Indiana, February 23, 2023.

Poster Presentation, “Building Trust in Automated Mobilities Through Engineering Design,” Design Theory & Methodology Student Poster Session, St. Louis, Missouri, August 23, 2022.

Panelist, “What I Wish I Would Have Known My First Year of Graduate School,” Purdue Graduate School, West Lafayette, Indiana, August 19, 2022.

Three Minute Thesis, “Let’s Be Excited for Shared Autonomous Vehicles,” Purdue University Campus Competition, West Lafayette, Indiana, April 12, 2022.

Invited Talk, “Fighting Insulin Resistance in America by Reinventing Last-Mile Delivery,” Presentation Given to NASA’s Convergent Aeronautics Solutions (CAS) Project Members, Virtual from West Lafayette, Indiana, May 10, 2021.

Poster Presentation, “The High-Tech, Global Design and Manufacturing Enterprise of 2035,” NSF/ASME Student Poster Session, Québec, Canada, August 27, 2018.

Teaching and Mentoring

Instructor of Record, Thermodynamics I (ME 20000), School of Mechanical Engineering, Purdue University. Taught, and currently teaching, a sophomore level engineering course in thermodynamics with an enrollment of 106 and 153 students, respectively. August 2024 – December 2024, August 2025 – December 2025.

Graduate Student Research Mentor, Summer Undergraduate Research Fellowship (SURF) Program, Purdue University. Mentored an undergraduate student from the University of Florida in functional near-infrared spectroscopy and how to employ it to accelerate the learning of novice drone pilots in a closed-loop context, West Lafayette, IN, May 2024 – July 2024.

Graduate Student Research Co-Leader, Summer Intensive Research Institute (SIRI), NSF Cyber Physical Systems Grant, Purdue University. Organized and directed an eight-week undergraduate research program with the graduate student leadership team. Performed several administrative duties, West Lafayette, Indiana, May 2023 – July 2023.

Seminary Co-Teacher, Seminaries & Institutes Division, Church Educational System, the Church of Jesus Christ of Latter-Day Saints. Taught an Old and New Testament course with my wife to high school students 5 days/week before school (50-minute lessons), West Lafayette, IN, August 2022 – May 2023.

Graduate Student Research Mentor, Summer Intensive Research Institute (SIRI), NSF Cyber Physical Systems Grant, Purdue University. Mentored two undergraduate students in the area of human-automation teaming, originally planned for West Lafayette, Indiana, but completed virtually due to COVID-19, June 2021 – July 2021.

Team Lead, Brigham Young University Capstone, Project for Deseret International Charities. Directed a multidisciplinary team of six engineering students through two stages of

product development, Provo, Utah, September 2019 – December 2019.

Teaching Assistant, Dynamics (CE EN 204), Civil Engineering Department, Brigham Young University. Taught and mentored students in the area of dynamics, Provo, Utah, May 2017 – April 2018.

Grader, Static Systems (ME EN 101), Mechanical Engineering Department, Brigham Young University. Graded homework and exams, Provo, Utah, May 2017 – June 2017.

Professional Activities

Journal Paper Reviews

Manuscript Reviewer, Journal of Mechanical Design, December 2025 - Present.

Manuscript Reviewer, Design Science, March 2025 - Present.

Conference Organization and Service

Judge, Spring Undergraduate Research Conference, West Lafayette, IN, April 10, 2025.

Judge, Summer Undergraduate Research Conference, West Lafayette, IN, July 25, 2024.

Co-Session Chair, Summer Intensive Research Institute (SIRI) Lightning Talks, West Lafayette, IN, July 24, 2024.

Judge, Summer Undergraduate Research Conference, West Lafayette, IN, July 27, 2023.

Session Chair, Summer Intensive Research Institute (SIRI) Lightning Talks, West Lafayette, IN, July 26, 2023.

Workshop Brainstorm Leader, Summit on Trusted Autonomy Research & Technology, West Lafayette, IN, June 28-29, 2022.

Manuscript Reviewer, ASME International Design Engineering Technical Conference (IDETC), 2019.

Professional Affiliations

Member, The Design Society, 2023 - Present.

Member, American Society of Mechanical Engineering (ASME), 2018 - Present.

Service

Committee Member, Grade Appeals Committee, College of Engineering, Purdue University, August 2023 – May 2024

President, Official Mechanical Engineering Graduate Association (OMEGA), Purdue University, May 2023 – April 2024.

Committee Representative, West Lafayette Mobility, May 2023 – December 2024.

Co-Vice President, Official Mechanical Engineering Graduate Association (OMEGA), Purdue University, May 2022 – April 2023.

Student Chair, American Society of Mechanical Engineers, Brigham Young University Chapter, September 2019 – April 2020.

Associate Editor, The BYU Design Review, July 2019 – June 2020.

Full-time Missionary, The Church of Jesus Christ of Latter-day Saints, Canada Montréal Mission. Served as Trainer (3x) and Zone Leader (2x), May 2014 – April 2016.

International Experience

Participant, 16th SIG Workshop on Design Theory and 7th SIG Tutorial on Design Theory, Paris, France, January – February 2023

Team Lead and Member, Brigham Young University Capstone, Sponsored by Deseret International Charities, travel to implement prototype in Ulaanbaatar, Mongolia planned, but ultimately cancelled due to COVID-19, March 2020

Intern, French National Institute of Health & Medical Research (INSERM), Paris, France, May – June 2019

Study Abroad Student, Global Product Development (Me En 579), USA, Panama, Brazil, Chile, and Argentina, April – May 2018

Full-time Missionary, the Church of Jesus Christ of Latter-day Saints, lived among and served people in the province of Québec, Canada, May 2014 – April 2016